Skin, nails and hair

20

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Introduction

The skin is the largest organ in the human body. Forming a major interface between man and his environment, it covers an area of approximately 2 m and weighs about 4 kg. The structure of human skin is complex (Figs 20.1 and 20.2), consisting of four distinct layers and tissue components with many important functions (Box 20.1). Reactions may occur in any of the components of human skin and their clinical manifestations reflect, among other factors, the skin level in which they occur, and sometimes they act as a 'window' of systemic changes elsewhere in the body, e.g. medical conditions discussed later in the chapter, such as those associated with pruritus (see Box 20.9), systemic causes of erythema nodosum (see Box 20.10) or paraneoplastic skin conditions (Box 20.2).

The accurate diagnosis of most skin lesions requires an adequate history, careful examination of the patient and, occasionally, laboratory investigation, but dermatology is predominantly a visual specialty.

History

Detailed information should be sought concerning the present skin condition. This should include the site of onset, mode of spread and duration of the disorder. Any personal history or family history of skin disease, including skin cancer and atopy (an allergic skin reaction becoming apparent more or less immediately on contact), is important. Previous medical conditions should be noted and a full drug history obtained, including the use of over-the-counter and other preparations. The social and occupational history and, in some circumstances, details of recent travel, environmental exposure, especially sunshine and artificial ultraviolet light, and sexual activity are often important (Box 20.3).

Examination

The whole skin, including hair, nails and assessable mucosae, should be fully inspected (preferably in natural light), but the patient's modesty should be protected. Sometimes a magnifying lens or dermatoscope is useful.

Colour and pigmentation

Before inspecting any rash or lesion, note the colour of the skin. Normal skin colour varies, depending on lifestyle and light exposure as well as constitutional and ethnic factors.

Pallor can have many causes. It may be:

- Temporary, owing to shock, haemorrhage or intense emotion
- Persistent, owing to anaemia or peripheral vasoconstriction

Vasoconstriction is seen in patients with severe atopy—an inherited susceptibility to asthma, eczema and hay fever. Pallor is a feature of anaemia, but not all pale persons are anaemic; conjunctival and mucosal colour is a better indication of anaemia than skin colour. A pale skin resulting from diminished pigment occurs with hypopituitarism and hypogonadism.

Normal skin contains varying amounts of brown melanin pigment. Brown pigmentation owing to deposited haemosiderin is always pathological. Albinism is an inherited generalized absence of pigment in the skin; a localized form is known as piebaldism. Patches of white and darkly pigmented skin (vitiligo) (Fig. 20.3) are owing to a local and complete absence of melanocytes. Several autoimmune endocrine disorders are associated with vitiligo.

Abnormal redness of the skin (erythema) is seen after overheating, extreme exertion and sunburn and in febrile, exanthematous and inflammatory skin diseases. Flushing is a striking redness, usually of the face and neck, which may be transient or persistent. Local redness may be caused by telangiectasia, especially on the face. Cyanosis is a blue or purple-blue tint caused by the presence of excessive reduced haemoglobin, either locally, as in impaired peripheral circulation, or generally, when oxygenation of the blood is defective. The skin colour in methaemoglobinaemia is more leaden than in ordinary cyanosis; it is caused by drugs, such as dapsone, and certain poisons.

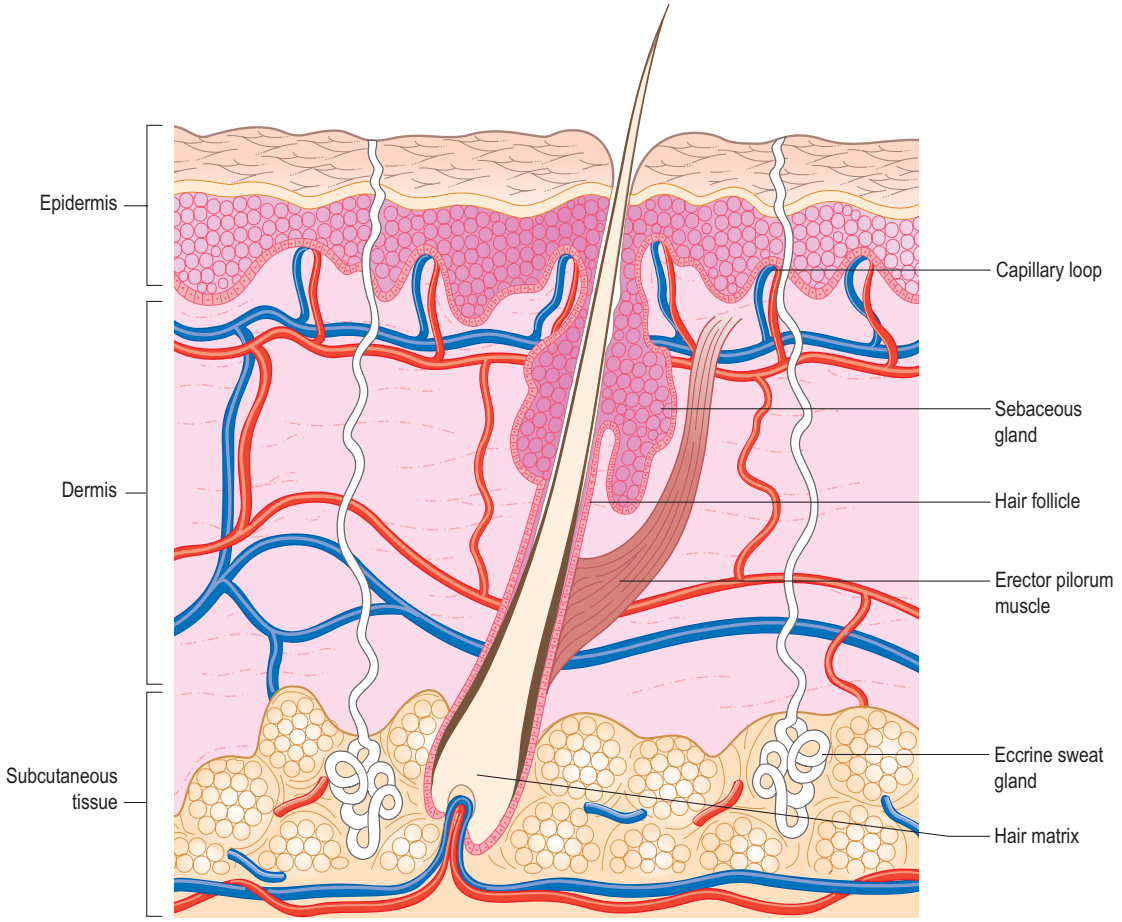


Figure 20.1 The anatomy of the full thickness of the skin in section.

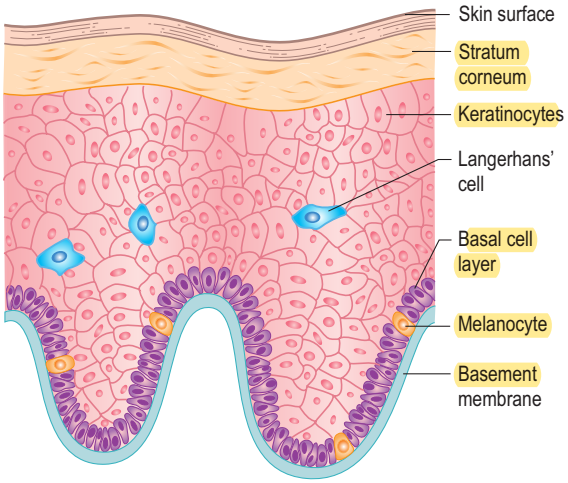


Figure 20.2 The anatomy of the epidermis.

Box 20.1 Functions of the human skin

- Protection: physical, chemical, infection—immune and innate
- Physiological: homoeostasis of electrolytes, water and protein
- Thermoregulation
- Sensation: specialized nerve endings—pain, touch and temperature
- Lubrication and waterproofing: sebum
- Immunological reactions: Langerhans' cells, lymphocytes, macrophages
- Wound healing
- Ultraviolet-induced vitamin D synthesis
- Body odour: apocrine glands
- Psychosocial: cosmetic

Jaundice varies from the subicteric lemon-yellow tints seen in pernicious anaemia and haemolytic jaundice to various shades of yellow, orange or dark olive-green in obstructive jaundice. Jaundice, which stains the conjunctivae, must be distinguished from the rare orange-yellow of carotenaemia, which does not. Slight degrees of jaundice cannot be seen in artificial light.

Increased pigmentation may be racial, owing to sunburn or connected with various diseases. In Addison's disease, a brown or dark-brown pigmentation affects exposed parts and parts not normally pigmented, such as the axillae and the

Box 20.2 Paraneoplastic skin disorders

Dermatosis	Associated tumour
Dermatomyositis	Lung, gastrointestinal (GI) tract, genitourinary tract
Acanthosis nigricans	GI tract, lung, liver
Paget's disease of the nipple/extra-mammary Paget's disease of the perineum	Adenocarcinoma
Erythroderma	Haematological
Tylosis-thickening of palms and soles	Oesophageal carcinoma
Ichthyosis	Lymphoma
Erythema gyratum repens	Lung, breast
Necrolytic migratory erythema	Glucagonoma



Figure 20.3 Vitiligo, a disorder of cutaneous pigmentation that is often autoimmune in origin and associated with other autoimmune disorders.

Box 20.3 Approach to dermatological patient history

- Time course of skin eruption
- Distribution of lesions including initially
- Symptoms: pruritus
- Family history: atopy and psoriasis
- Drug/allergy history
- Past medical history
- Contacts: family and partners
- Provocating factors: sunlight and foods
- Previous and current treatments



Figure 20.4 Flat-topped papules of lichen planus.

palmar creases; the lips and mouth may exhibit dark bluish-black areas. Note, however, that mucosal pigmentation is a normal finding in a substantial proportion of black patients.

More or less generalized pigmentation may also be seen in the following:

- Haemochromatosis, in which the skin has a peculiar greyish-bronze colour with a metallic sheen, owing to excessive melanin and iron pigment
- Chronic arsenic poisoning, in which the skin is finely dappled affecting covered more than exposed parts
- Argyria, in which the deposition of silver in the skin produces a diffuse slate-grey hue
- The cachexia of advanced malignant disease

In pregnancy, there may be pigmentation of the nipples and areolae, of the linea alba and sometimes a mask-like pigmentation of the face (chloasma). Chloasma may also be induced by oral contraceptives containing oestrogen. A similar condition, melasma, may be seen in Asian and Afro-Caribbean males.

Localized pigmentation may be seen in pellagra and in scars of various kinds, particularly those owing to X-irradiation therapy. Venous hypertension in the legs is often associated with chronic purpura, leading to haemosiderin pigmentation. The mixture of punctate and fresh purpura and haemosiderin may produce a golden hue on the lower calves and shins. Pigmentation may also occur with chronic infestation by body lice. Erythema ab igne, a reticular pattern of pigmentation, can be seen in patients who use local heat to relieve chronic pain or on the shins of people who habitually sit too near a fire. Livedo reticularis, a web-like pattern of reddish-blue discolouration mostly involving the legs, occurs in autoimmune vasculitis, especially in systemic lupus erythematosus (SLE) and antiphospholipid syndrome, when it is associated with cerebral infarction. The violet-coloured lesions of lichen planus are slightly raised, flat-topped papules (Fig. 20.4). Psoriasis usually presents as a symmetrical plaque on extensor surfaces (Fig. 20.5). Keloid consists of raised and inflamed, overgrown tender scar tissue (Fig. 20.6). Dermatomyositis often produces swelling and heliotrope-coloured erythema of the eyelids without scaling of the skin.



Figure 20.5 Salmon-pink plaque of psoriasis on elbow covered in characteristic silver scale.



Figure 20.6 Multiple keloid scarring of the back caused by acne vulgaris. There is a genetic predisposition to the formation of keloid in scar tissue.

Skin lesions and eruptions

Skin eruptions and lesions should be examined with special reference to their morphology, distribution and arrangement. The terminology of skin lesions is summarized in Boxes 20.4 and 20.5. Colour, size, consistency, configuration, margination and surface characteristics should be noted.

Morphology of skin lesions

Inspection and palpation

Assessment of morphology requires visual and tactile examination. Do not be afraid to feel the lesions. You will rarely be exposed to any infection risk, with the exception of herpes simplex, herpes zoster, syphilis, hepatitis B and human immunodeficiency virus (HIV) disease. If these infections are suspected, it is wise to wear disposable plastic gloves when examining open or bleeding cutaneous lesions. Begin with palpation of the skin. Pass the hand gently over it, pinching it up between the forefinger and thumb, and note the following points:

- Is it smooth or rough, thin or thick?
- Is it dry or moist?
- Is there any visible sweating, either general or local?

The elasticity of the skin should be investigated. If a fold of healthy skin is pinched up, it immediately flattens itself out again when released. Sometimes, however, it only does so very slowly, remaining creased for a considerable time. This is found frequently in healthy old people, but may be an important sign of dehydration, for example after severe vomiting and diarrhoea, or in uncontrolled diabetes mellitus.

Subcutaneous oedema

When oedema is present, firm pressure on the skin with a finger produces a shallow pit that persists for some time. In some cases, especially when the oedema is very long standing, pitting is not found. The best place to look for slight degrees of oedema in cardiac disease is behind the malleoli at the ankles in patients who are ambulant and over the sacrum in

Box 20.4 Primary skin lesions: a glossary of dermatological terms

Macule	Non-palpable area of altered colour
Papule	Palpable elevated small area of skin (<0.5 cm)
Plaque	Palpable flat-topped discoid lesion (>2 cm)
Nodule	Solid palpable lesion within the skin (>0.5 cm)
Papilloma	Pedunculated lesion projecting from the skin
Vesicle	Small fluid-filled blister (<0.5 cm)
Bulla	Large fluid-filled blister (>0.5 cm)
Pustule	Blister containing pus
Wheal	Elevated lesion, often white with red margin owing to dermal oedema
Telangiectasia	Dilatation of superficial blood vessel
Petechiae	Pinhead-sized macules of blood
Purpura	Larger petechiae that do not blanch on pressure
Ecchymosis	Large extravasation of blood in skin (bruise)
Haematoma	Swelling owing to gross bleeding
Poikiloderma	Atrophy, reticulate hyperpigmentation and telangiectasia
Erythema	Redness of the skin
Burrow	Linear or curved elevations of the superficial skin owing to infestation by female scabies mite
Comedo	Dark horny keratin and sebaceous plugs within pilosebaceous openings

Box 20.5	Secondary skin lesions that evolve from primary lesions
Scale	Loose excess normal and abnormal horny layer
Crust	Dried exudate
Excoriation	A scratch
Lichenification	Thickening of the epidermis with exaggerated skin margin
Fissure	Slit in the skin
Erosion	Partial loss of epidermis which heals without scarring
Ulcer	At least the full thickness of the epidermis is lost. Healing occurs with scarring
Sinus	A cavity or channel that allows the escape of fluid or pus
Scar	Healing by replacement with fibrous tissue
Keloid scar	Excessive scar formation (see Fig. 20.6)
Atrophy	Thinning of the skin owing to shrinkage of epidermis, dermis or subcutaneous fat
Stria	Atrophic pink or white linear lesion owing to changes in connective tissue

those who are confined to bed. The finger pressure should be maintained for 20–30 seconds or slight degrees of oedema will be overlooked. Pitting is minimal or absent in oedema owing to lymphatic obstruction, where the skin is usually thickened and tough.

Subcutaneous emphysema

Air trapped under the skin gives rise to a characteristic crackling sensation on palpation. It starts in, and is usually confined to, the neighbourhood of the air passages. On rare occasions, it may result from the clostridial infection of soft tissues after injury (gas gangrene).

Distribution of skin lesions

Consider the distribution of an eruption by looking at the whole skin surface:

- Is it symmetrical or asymmetrical? Symmetry often implies an internal causation, whereas asymmetry may imply external factors.
- Is the eruption centrifugal (radiating from the centre) or centripetal (radiating to the centre)? Certain common diseases, such as chickenpox and pityriasis rosea, are characteristically centripetal, whereas erythema multiforme and erythema nodosum are centrifugal. Smallpox, now eradicated, was also centrifugal.

? A disease may exhibit a flexor or an extensor bias in its distribution: atopic eczema in childhood is characteristically flexor, whereas psoriasis in adults tends to be extensor.



Figure 20.7 Hansen's disease. A depigmented area of anaesthetic and slightly pink skin is seen on the exposed cheek. In this lepromatous lesion, acid-fast bacilli were found in scrapings.

- Are only exposed areas affected, implicating sunlight or some other external causative factor?
- If sunlight is suspected, are areas normally in shadow involved?
- Are the genitalia involved?
- Localized distributions may point immediately to an external contact as the cause, for example contact dermatitis from nickel earrings, lipstick dermatitis, etc.

Swelling of the eyelids is an important sign. Without redness and scaling, bilateral periorbital oedema may indicate acute nephritis, nephrosis or trichinosis. If there is irritation, contact dermatitis is the probable diagnosis. Dermatomyositis often produces swelling and heliotrope-coloured erythema of the eyelids without scaling of the skin. In Hansen's disease (leprosy), the skin lesions may be depigmented or reddened, with a slightly raised edge; they are also anaesthetic to pinprick testing (Fig. 20.7) and mainly located in skin that is normally cooler than core body temperature.

Configuration of skin lesions

Once the morphology of individual lesions and their distribution has been established, it is useful to describe their configuration on the skin (Box 20.6).

The hair

Hair colour and texture are racial characteristics that are genetically determined. The yellow-brown Mongol race has black straight hair, negroid people have black curly hair and Caucasians have fair, brown, red or black hair. Secondary sexual hair begins to appear at puberty and has characteristic male and female patterns. Androgenic male pattern baldness is genetically determined, but requires adequate levels of circulating androgens for its expression. It occurs in women only in old age.

Growth of hair

Unlike other epithelial mitotic activity that is continuous throughout life, the growth of hair is cyclic (Fig. 20.8), the hair follicle going through alternating phases of growth (anagen) and rest (telogen). Anagen in the scalp lasts 3–5 years; telogen is much shorter, about 3 months. Catagen is the conversion stage from active to resting and it usually lasts a few days. The duration of the anagen phase determines the length to which hair in different body areas can grow. On the scalp, there are on average about 100 000 hairs. The normal scalp may shed as many as 100 hairs every day as a normal consequence of growth cycling. These proportions can be estimated by looking at plucked hairs (trichogram); the 'root' of a telogen hair is non-pigmented and visible as a white, club-like swelling. Normally 85% of scalp hairs are in anagen and 15% in telogen.

Box 20.6 Configuration of individual lesions

- Nummular/discoid
- Round or coin-like
- Annular
- Ring-like
- Circinate
- Circular
- Arcuate
- Curved
- Gyrate/serpiginous
- Wave-like
- Linear
- In a line
- Grouped
- Clustered
- Reticulate
- Net-like

Alopecia

Hair loss (alopecia) has many causes. It is convenient to subdivide alopecia into localized and diffuse types. In addition, the clinician should determine whether the alopecia results in scarring and hence, permanent hair loss (Box 20.7).

Any inflammatory or destructive disease of the scalp skin may destroy hair follicles in its wake. Thus, burns, heavy X-ray irradiation or herpes zoster infection in the first division of the trigeminal nerve may cause scarring and alopecia. Alopecia in the presence of normal scalp skin may be patchy and localized, as in traction alopecia in nervous children, ringworm infections (tinea capitis) or autoimmune alopecia areata. Secondary syphilis is a rare cause of a patchy alopecia with a 'moth-eaten' appearance.

Scalp hair loss at the temples and crown, with the growth of male-type body hair, is characteristic of women with virilizing disorders. Metabolic causes of diffuse hair loss in women include hypothyroidism and severe iron deficiency anaemia. Antimitotic drugs may affect the growing hair follicles, producing a diffuse loss of anagen hairs, which are pigmented throughout their length. Dramatic metabolic upsets, such as childbirth, starvation and severe toxic illnesses, may precipitate follicles into the resting phase, producing an effluvium of telogen hairs 3 months later, when anagen begins again. This is called telogen alopecia. In the autoimmune disorder alopecia totalis (Fig. 20.9), there is complete loss of hair. Self-inflicted traction alopecia (Fig. 20.10) may indicate psychological problems.

The nails

The nails should be examined carefully. The structure of the nail and nail bed is shown in Figs 20.11 and 20.12. The nail consists of a strong,

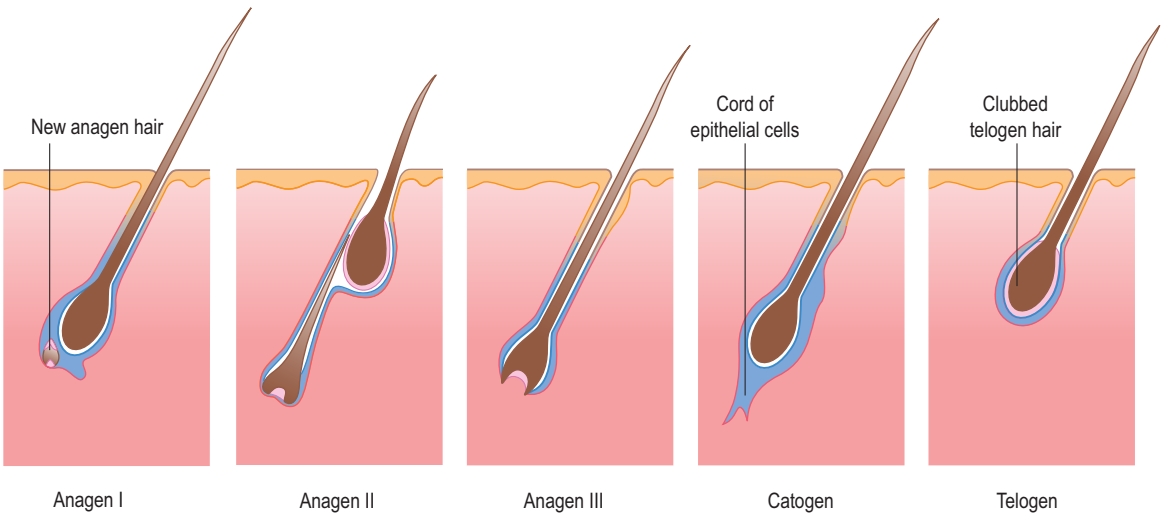


Figure 20.8 Hair follicle growth stages.

Box 20.7 Causes of alopecia

Non-scarring

- Alopecia areata
- Trichotillomania (self-induced hair pulling)
- Traction alopecia
- Scalp ringworm (human)
- Metabolic iron deficiency, hypothyroidism
- Drugs (e.g. heparin, retinoids, chemotherapy)
- Sarcoidosis

Scarring

- Burns, trauma
- Aplasia cutis
- Discoid lupus erythematosus, lichen plakophilins
- Dissecting folliculitis
- Pseudopelade
- Kerion (see Fig. 20.30)
- X-irradiation
- Necrobiosis



Figure 20.10 Traction alopecia in a psychologically troubled patient.

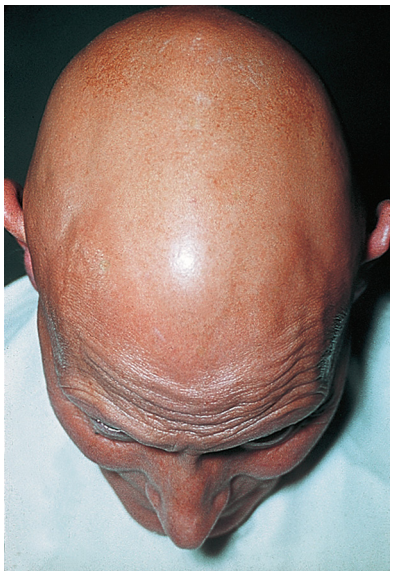


Figure 20.9 Alopecia totalis.

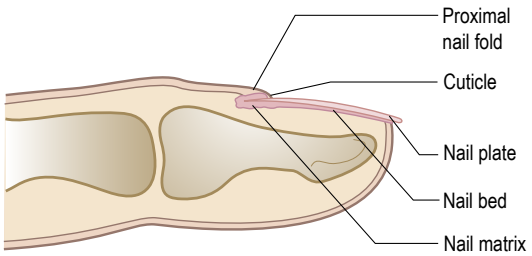


Figure 20.11 Structure of the nail (lateral view).

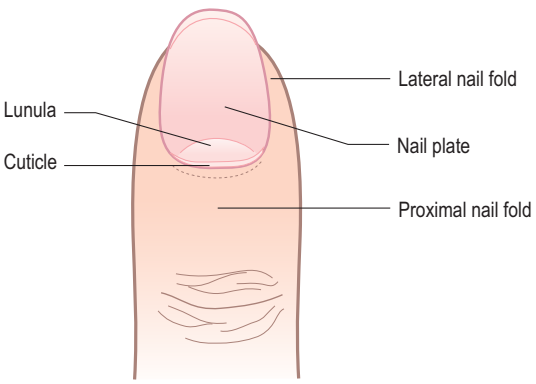


Figure 20.12 Structure of the nail (dorsal view).

relatively inflexible keratinous nail plate over the dorsal surface of the end of each digit, protecting the fingertip.

Nail matrix abnormalities

Thimble pitting of the nails is characteristic of psoriasis (Fig. 20.13), but eczema and alopecia areata may also produce pitting. A severe illness may temporarily arrest nail growth; when growth starts again, transverse ridges develop. These are called Beau's lines and can be used to date the time of onset of an illness. Inflammation of the cuticle or nail fold (chronic paronychia) may produce similar changes. The changes described above arise from disturbance of the nail matrix.

Nail and nail-bed abnormalities

Disturbance of the nail bed may produce thick nails (pachyonychia) or separation of the nail from the bed (onycholysis). This occurs in psoriasis, but may be idiopathic. Long-term tetracyclines may induce separation when the fingers are exposed to strong sunlight (photo-onycholysis). The nail may be destroyed in severe lichen planus (Fig. 20.14) or epidermolysis bullosa (a genetic abnormality in which the skin blisters in response to minor trauma). Nails



Figure 20.13 Psoriasis of the nail beds.



Figure 20.14 Lichen planus, showing longitudinal ridging of the nails and overgrowth of the cuticle on the nail plate (pterygium).

are missing in the inherited nail-patella syndrome. Splinter haemorrhages under the nails may result from trauma, psoriasis, rheumatoid arthritis or other 'collagen vascular' diseases, bacterial endocarditis and trichinosis.

The nails in systemic disease

In long-lasting iron-deficiency states, the fingernails and toenails become soft, thin, brittle and spoon-shaped. They lose their normal transverse convex curvature, becoming flattened or concave (koilonychia). The 'half and half nail', with a white proximal and red or brown distal half, is seen in some patients with chronic renal failure. Whitening of the nail plates may be related to hypoalbuminaemia, as in cirrhosis of the liver (leukonychia). Some drugs, notably antimalarials, antibiotics and phenothiazines, may discolour the nail. Nail-fold telangiectasia or erythema is a useful physical sign in dermatomyositis (Fig. 20.15), systemic sclerosis and systemic lupus erythematosus. In dermatomyositis, the cuticle becomes ragged. In systemic sclerosis, loss of finger pads may lead to curvature of the nail plates. An impaired peripheral circulation, as in Raynaud's phenomenon, can lead to thinning and longitudinal ridging of the nail plate, sometimes with partial onycholysis. In bronchiectasis, the nails may take on a curved, yellow appearance (Fig. 20.16).



Figure 20.15 Typical heliotrope-coloured erythema seen in dermatomyositis (periungual and Gottron's papules over the knuckles).



Figure 20.16 Yellow nail syndrome in bronchiectasis.

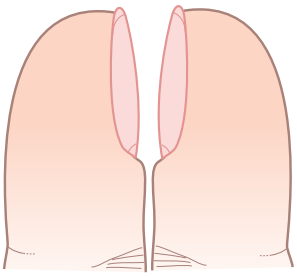
Clubbing

Clubbing is probably caused by hypervascularity and the opening of anastomotic channels in the nail bed. Rarely, clubbing may be congenital. The distal end of the digit becomes expanded, with the nail curved excessively in both longitudinal and transverse planes. Viewed from the side, the angle at the nail plate is lost and may exceed 180°. In normal nails, when both thumbnails are placed in apposition, there is a lozenge-shaped gap whereas, in clubbing, there is a reduction in this gap (Schamroth's window test; Fig. 20.17). In hypertrophic pulmonary osteoarthropathy, there is clubbing of the fingers and thickening of the periosteum of the radius, ulna, tibia and fibula, which can be tender. (The causes of clubbing are listed in Box 20.8.)

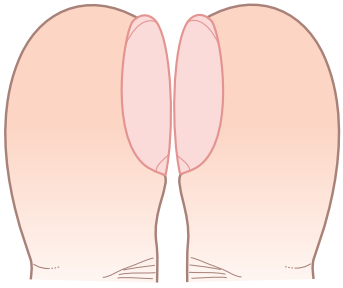
Cutaneous manifestations of internal disease

Genodermatoses (lesions of inherited origin)

White macules shaped like small ash leaves and present at birth may be the first sign of tuberous



Normal nail



Clubbed nail

Figure 20.17 Schamroth's window test.

Box 20.8 Causes of clubbing of the fingers

Cardiopulmonary disorders

- Severe chronic cyanosis
- Cyanotic congenital heart disease
- Chronic fibrosing alveolitis
- Chronic suppuration in the lungs:
  - —Bonchiectasis
  - —Empyema
  - —Lung abscess
- Carcinoma of the bronchus
- Bacterial endocarditis

Chronic abdominal disorders

- Crohn's disease
- Ulcerative colitis
- Cirrhosis of the liver

sclerosis. Sometimes they are difficult to see by natural light, but show up under ultraviolet light (Wood's light). They should be looked for in infants with seizures. Pigmentation of the lips is a feature of the genetically determined Peutz-Jeghers syndrome, in which multiple polyps of the stomach or colon appear that may later undergo malignant transformation. Café au lait type macules, sometimes multiple, are a common sign of neurofibromatosis (NF). Another valuable sign in von Recklinghausen's neurofibromatosis is bilateral freckling of the axillary skin. The characteristic soft neurofibromata may



Figure 20.18 Discoid lupus erythematosus, showing atrophic scars.

be solitary or a few or hundreds may be scattered over the body. Ichthyosis (scaly fish skin) is usually present from childhood and is genetic. If ichthyosis is acquired in adult life, a search should be made for malignancy or other underlying disease.

Non-organ-specific autoimmune disorders

Several of the so-called collagen vascular diseases show characteristic cutaneous eruptions. Systemic lupus erythematosus, seen in women between puberty and the menopause, may show a symmetrical 'butterfly' erythema of the nose and cheeks. In discoid lupus, the cutaneous lesion is localized (Fig. 20.18). In polyarteritis nodosa, reticular livedo of the limbs with purpura, vasculitic papules and ulceration occurs. In scleroderma (systemic sclerosis), acrosclerosis of the fingertips with scarring, ulceration and calcinosis follows a Raynaud's phenomenon of increasing severity. Dermatomyositis often presents with a heliotrope-coloured discolouration and oedema of the eyelids and with fixed erythema over the dorsa of the knuckles and fingers (see Fig. 20.15) and over the bony points of the shoulders, elbows and legs. There is usually weakness of the proximal limb muscles. Dermatomyositis in the middle-aged is associated with internal malignancy in about 10% of cases.

Skin pigmentation

Acanthosis nigricans is a brownish velvety thickening of the axillae, groins and sides of the neck. Sometimes there is thickening of the palms and soles, and warty excrescences may develop on the skin, eyelids and oral mucosa. In the middle-aged, acanthosis nigricans is strongly associated with internal malignancy, but benign minor forms are seen in obese young women, especially in Arab people, and in children with endocrinopathies characterized by insulin resistance. Patchy depigmented macules which are hypoanaesthetic and associated with enlargement of the peripheral nerves are a feature of certain types of leprosy (Hansen's disease).

Generalized, severe persistent pruritus in the absence of obvious skin disease may be caused by systemic disease (Box 20.9). However, in older people with dry skin, it is common and of no systemic significance. Diabetes mellitus has a number of skin manifestations; of these, pruritus vulvae, pruritus ani, balanoposthitis and angular stomatitis are caused by *Candida* overgrowth. Boils, follicular pustules or ecthyma are staphylococcal in origin. Impetigo and erysipelas, both streptococcal infections, are uncommon (Fig. 20.19). Eruptive xanthomata are a rare feature of uncontrolled diabetes mellitus. Necrobiosis lipoidica diabeticorum (Fig. 20.20) produces reddish-brown plaques, usually on the shins, with central atrophy of the skin. It has to be distinguished from peritibial myxoedema (which is hypertrophic, not atrophic), the dermatoliposclerosis of chronic venous disease in the legs and the epidermal hypertrophy of chronic lymphatic obstruction.

Neuropathic ('perforating') ulcers, which are found on pressure points of the heel, ball of the foot or toes, are characteristically painless. Arteriopathic ulceration resulting from large vessel disease is seen on the foot and on the calves in small vessel disease.

Box 20.9 Causes of pruritus in systemic disease

- Iron deficiency anaemia
- Diabetes mellitus
- Thyrotoxicosis or hypothyroidism
- Chronic renal failure
- Chronic hepatic failure
- Biliary obstruction, including primary biliary cholangitis
- Lymphoma and other internal malignancies
- Drugs (e.g. cocaine, morphine)
- HIV infection
- Polycythemia vera
- Parasites (e.g. onchocerciasis)
- Psychogenic

Spider naevi consist of a central arteriole feeding a cluster of surrounding vessels. Many young people have up to seven naevi on the face, shoulders or arms. In others, pregnancy, the administration of oestrogens (as in oral contraceptives) and liver disease may cause multiple lesions. Pregnancy and liver disease may also produce a blotchy erythema of the thenar and hypothenar eminences ('liver palms'). Leuconychia are seen in liver disease.

Erythema nodosum is a condition in which tender, painful, red nodules appear, typically on the shins. They fade slowly over several weeks, leaving bruising, but never ulcerate. Sarcoidosis, inflammatory bowel disease and drug sensitivity are the most common causes, but other systemic disorders should be considered (Box 20.10).

Xanthomata are yellow or orange papules or nodules in the skin caused by dermal aggregations of lipid-loaded cells. Different patterns of hyperlipoproteinaemia may induce varying patterns of xanthomatosis. Thus, the type Ia (hypercholesterolaemia) pattern typically causes tuberous xanthomata on the extensor aspects of the knees and elbows and on the buttocks, sometimes associated with tendon xanthomata. Widespread eruptive xanthomata are more characteristic of hypertriglyceridaemia. White deposits of lipid (arcus senilis) in the cornea may have a similar explanation, but may be a normal feature in those over 60. Flat lipid deposits around the eyes (xanthelasma) may be caused by hyperlipidaemia, but they are also seen in the middle-aged and elderly without any general metabolic upset.

Carotenaemia produces an orange-yellow colour to the skin, especially of the palms and soles. It occurs in those who eat great quantities of carrots and other vegetables, in hypothyroidism, in diabetic patients and also in those taking β-carotene for the treatment of porphyria.



Figure 20.19 **(A)** Impetigo. **(B)** Erysipelas.



Figure 20.20 Necrobiosis lipoidica diabeticorum.

Box 20.10 Causes of erythema nodosum

- Sarcoidosis
- Drugs (e.g. sulphonamides)
- Streptococcal infection
- Tuberculosis
- Inflammatory bowel disease
- Behçet's disease
- Other infections (e.g. leprosy, systemic mycoses, toxoplasmosis, lymphogranuloma venereum)

Haemorrhage in the skin

Aggregations of extravasated red blood cells in the skin cause purpura (Fig. 20.21). Purpura may punctate from capillary haemorrhage or may form larger macules, depending on the extent of haemorrhage and the size of vessels involved. Palpable purpura should raise suspicion of leucocytoclastic vasculitis. The Hess test for capillary fragility involves deliberately inducing punctate purpura on the forearm by inflating a cuff above the elbow at 100 mmHg for 3 minutes. Sensitivity to drugs such as aspirin may cause widespread 'capillaritis'.

The term ecchymosis implies a bruise, usually with cutaneous and subcutaneous haemorrhage causing a palpable lump. A frank fluctuant collection of blood is a haematoma. Unlike erythema and telangiectasia, purpura cannot be blanched by pressure. It must not be confused with senile haemangioma (cherry angioma or Campbell de Morgan spot), which is common in later life on the trunk and has no pathological significance. Haemorrhage into the thick epidermis of the palm or sole caused by trauma (e.g. 'jogger's heel') may induce brown or almost black macules that take weeks to disappear, inviting confusion with melanoma.

The skin in sexually transmitted diseases

The skin is involved in various sexually transmitted diseases. The primary chancre of syphilis may occur on the genitalia of either sex, at or near the anus, on the lip or, rarely, elsewhere. The rash of secondary syphilis is brownish-red and maculopapular; it is one of the few rashes involving the palms and soles. It does not itch. Other manifestations of the secondary stage are condylomata lata around the anogenital area and snail-track ulceration and 'mucous patches' in the mouth. There may be low-grade fever, lymphadenopathy and splenomegaly.



Figure 20.21 Purpura in Henoch-Schönlein disease.

Septicaemia is a rare complication of gonorrhoea that occurs particularly in pregnant women presenting with pustular skin lesions. Recurrent type II herpes simplex is common on the penis; it occurs less often on the buttocks, where relapses may be heralded by a radiating neuralgia. Genital viral warts are common in both sexes. They are particularly important in females because evidence indicates that certain viral subtypes are responsible for chronic cervical dysplasia with malignant potential.

HIV-related syndromes have many cutaneous manifestations, including disseminated Kaposi's sarcoma, candidiasis, molluscum contagiosum, seborrhoeic dermatitis, folliculitis and oral hairy leukoplakia (Fig. 20.22).

Viral infection of the skin

Several of the most common viral infections and illnesses of childhood are characterized by fever and a distinctive rash (exanthem), including measles, varicella and rubella. In measles, upper respiratory symptoms are quickly followed by a characteristic maculopapular and erythematous rash. In rubella (German measles), the rash is more transient, with small papules, and associated



Figure 20.22 Hairy leukoplakic.



Figure 20.23 Herpes simplex (type I).



Figure 20.24 Herpes zoster vesicles on the ear lobe involving the C2 dermatome.

with occipital lymphadenopathy and only slight malaise. The exanthem of varicella (chickenpox) is papulovesicular and centripetal, and there may be lesions in the mouth. In herpes zoster and herpes simplex infections (Fig. 20.23), there is a non-follicular papulovesicular rash in which the vesicles are planted in an inflamed base. The rash is painful. In herpes zoster, the rash follows a segmental distribution, corresponding with skin of a dermatome. The vesicular lesion becomes encrusted (Fig. 20.24) and later, secondary infection may occur.

Other less common viral infections associated with a rash include erythema infectiosum, caused by a parvovirus, in which the exanthem on the face gives a 'slapped-cheek' appearance, and roseola infantum, a disease of toddlers which mimics rubella.

Drug eruptions

In the last 40 years, eruptions caused by drugs have become common. Most such rashes are caused by an allergic hypersensitivity. Drug rashes can mimic almost every pattern of skin disease. Thus, urticaria may be caused by penicillin or opiates; a measles-like (morbilliform) rash may be induced by ampicillin, especially when given to patients with infectious mononucleosis; eczema-like rashes are seen with methyldopa and phenylbutazone therapy; and gold and chloroquine rashes mimic lichen planus. A palpable rash looking like Henoch-Schönlein purpura indicates leucocytoclastic vasculitis. Generalized exfoliative dermatitis may be induced by sulfonylureas, indomethacin and allopurinol. Increasing use of biologics in clinical medicine has identified unusual skin eruptions.

Drugs that may sensitize the skin to sunlight (phototoxic reaction) include tetracyclines, sulfonamides and nalidixic acid. Acne-like rashes may follow high-dose prednisolone therapy and are common with phenytoin therapy. Certain cytotoxic drugs and sodium valproate cause hair loss. Both erythema nodosum and erythema multiforme may be induced by sulfonamides, including co-trimoxazole. Laboratory tests are of little value in the diagnosis of drug eruptions. A carefully taken history and knowledge of the common patterns of drug reactions usually allow accurate diagnosis (Box 20.11).

Tumours in the skin

Exposure to the sun may, after many years, result in the development of many common skin tumours, for example squamous or basal cell carcinoma or melanoma. These tumours are especially common in fair-skinned people. Basal cell carcinoma arises especially on the face, near the nose or on the forehead (Fig. 20.25). The lesion may be ulcerated with a firm, rounded edge, or papular. Melanomas

Box 20.11 Common drug eruptions

Morphological type	Drug
Maculopapular	Penicillin/amoxicillin
Urticaria	Penicillin, non-steroidal drugs
Vasculitis	Hydralazine
Fixed drug eruption	Tetracyclines, phenolphthalein
Pigmentation	Amiodarone, minocycline
Photosensitivity	Thiazides, sulfonamides
Lupus erythematosus	Isoniazid, penicillamine
Erythema nodosum	Oral contraceptive pill, sulfonamides
Erythema multiforme	Barbiturates
Acneform	Steroids
Pustular	Carbamazepine
Lichenoid	Thiazides, antimalarials, allopurinol
Psoriasiform	Beta-blockers, lithium, antimalarials
Toxic epidermal necrolysis/ Stevens-Johnson syndrome/ erythema multiforme	Carbamazepine, non-steroidal drugs, co-trimoxazole
Pemphigus	Angiotensin converting enzyme (ACE) inhibitors
Linear IgA disease	Vancomycin
Erythroderma	Sulfonylureas, ACE inhibitors, allopurinol

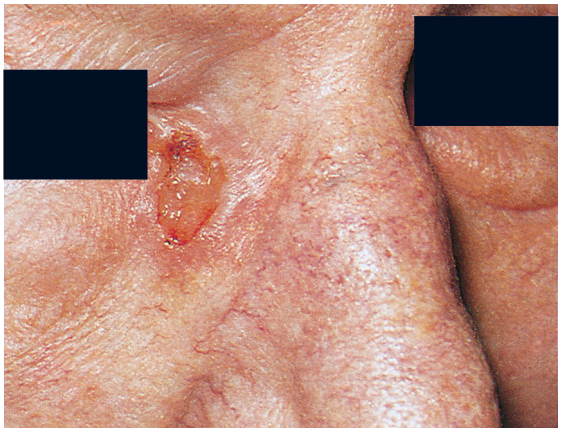


Figure 20.25 Basal cell carcinoma.

may develop on the torso in men and legs in females. They may be pigmented or amelanotic and may in about a third of cases develop rapidly in a pre-existing benign mole (Fig. 20.26). Staging by assessing draining regional sentinel lymph nodes is becoming increasingly routine in cutaneous oncology. Seborrhoeic keratoses are familial, resulting in a raised warty pigmented lesions found in the sun-exposed elderly, especially on the dorsa of hands as 'liver spots' (Fig. 20.27). A symptomatic pigmented skin lesion having an asymmetric, ragged border, three or more colours and diameter of more than 7



Figure 20.26 Malignant melanoma on a male chest. The nodule is invasive.



Figure 20.27 A pigmented basal cell papilloma (seborrhoeic keratosis) on the face. This is a benign lesion.

mm, particularly if it is increasing in size or bleeding and found on sun-damaged freckled skin, should be regarded as suspicious of malignant melanoma.

Special techniques in examination of the skin

The skin is uniquely available to the examining physician. There are a number of diagnostic procedures and blood tests (Box 20.12).

Tzanck preparation

Tzanck preparation, an often forgotten old but valuable technique, is useful for emergency diagnosis of vesicular infections or blistering eruptions, such as pemphigus. The intact blister is opened and the base gently scraped. The material obtained is smeared onto the microscope slide, allowed to air dry and then stained. Viral lesions will show typical multinucleated giant cells, and pemphigus will show acantholytic cells.

Microscopic examination—dermatoscopy (direct assessment of skin lesions)

Dermatoscopic and/or microscopic examination is useful in the diagnosis of scabies, pediculosis (lice)

and fungal infection (tinea and candidiasis). The established routine use of the dermatoscope has become essential in the assessment of pigmented skin lesions and aids in diagnosis of skin malignancy, especially malignant melanoma.

Scabies

Scabies is caused by the mite *Acarus (Sarcoptes) scabei*. The female is larger than the male and burrows in the epidermis, depositing eggs. These burrows should be looked for between the fingers, on the hands or wrists and on the sides of the feet. They can be recognized with the naked eye as short dark lines terminating in a shining spot of skin. The eggs lie in the dark line, the mite in the shining spot. It may be picked out by means of a flat surgical needle and placed on a microscope slide for more detailed examination.

Pediculosis

Three forms of pediculosis or louse infestation occur: *Pediculus capitis* on the head, *Pediculus corporis* on the trunk and *Pediculus pubis* on the pubic and axillary hairs. The eggs or nits of *P. pubis* and *P. capitis* adhere to the hairs. From their position on the hairs, one can judge roughly the duration of the condition: they are fixed at first near the root of the hair and are carried up as the hair grows, so the higher the nits are, the longer the pediculi have been present. *P. corporis* should be looked for in the seams of the clothes, especially where the clothes come into contact with the skin, for example over the shoulders. The bites of the parasite produce haemorrhagic spots, each with a dark centre and a paler areola. Marks of scratching should be looked for on parts accessible to the patient's nails. *P. pubis* is venereally acquired and causes intense pubic itching. The nits are laid on the pubic hair and the lice themselves are easily visible. *P. corporis* is seen only in the grossly deprived, living in rough conditions of war or social upheaval. In contrast, *P. capitis* is common in school children, however clean they and their families may be, and is endemic in many schools.

Box 20.12 Dermatology blood investigations

Blood test	Clinical example
Full blood count (FBC)	Pruritus
Ferritin and iron studies	Hail and nail
Biochemistry renal and liver function	Pruritus
Streptococcal serology	Cellulitis
Autoantibodies	Autoimmune disorders
Hepatitis screen	Lichen planus
HSV 1 and 2 serology	Erythema multiforme
Syphilis serology	Secondary syphilis
HIV	Persistent infections
HLA typing	Dermatitis herpetiformis
DNA analysis	Epidermolysis bullosa

Fungal infections

Fungus may grow in the skin, nails or hair and can cause disease (ringworm or tinea), for example athlete's foot (tinea pedis).

The most common sites of fungal infection are the skin between the toes (tinea pedis) and the soles of the feet and the groin (tinea cruris). The lesions may be scaly or vesicular, tending to spread in a ring form with central healing (Fig. 20.28); macerated, dead, white, offensive-smelling epithelium is found in the intertriginous areas, such as the toe clefts.

Nail discolouration, deformity, hypertrophy and abnormal brittleness may result from fungus infection (tinea unguium).

Ringworm of the scalp (tinea capitis) is most common in children. It presents as round or oval areas of baldness covered with short, lustreless broken-off hair stumps. These hair stumps may fluoresce bright green under Wood's light. Some fungi do not fluoresce with Wood's light, however, and these can be detected only by microscopy and culture (Figs 20.29 and 20.30).

Microscopic examination for fungus infection

Scales from the active edge of a lesion are scraped off lightly with a scalpel or the roofs of vesicles are snipped off with scissors. The material is placed in a drop of 10% to 20% aqueous potassium hydroxide solution on a microscope slide, covered with a coverslip and left for 30 minutes to clear. It is then examined under the light microscope with the 8-mm or 4-mm objective using low illumination. The mycelia are recognized as branching, refractile threads that boldly transgress the outlines of the squamous cells. Nails are examined in much the same way, but it is necessary to break up the snippings



Figure 20.28 Tinea rubrum (ringworm infection).



Figure 20.29 Lactophenol blue preparation showing macronidia of Microsporum species, isolated from skin scrapings from a patient with ringworm.



Figure 20.30 Kerion owing to localized deep dermatophyte infection in the scalp.

and shavings into small fragments. These are either heated in potassium hydroxide or are left to clear in it overnight before being examined.

A scalp lesion is cleaned with 70% alcohol or with 1% cetrimide; infected stumps and scales are removed by scraping with a scalpel. The hairs are cleaned in potassium hydroxide in the same way as skin scales. Examination under the microscope reveals spores on the outside of the hair roots and mycelia inside the hair substance. The species of fungus responsible may be established by culture on Sabouraud's glucose-agar.

Wood's light

A Wood's light lamp emits long-wave ultraviolet light at a peak of 360 nm. Wood's light examination is performed in a darkened room and is useful to identify the fluorescence of fungi and corynebacterial infection (erythrasma), elevated porphyrins in urine or in the localization of pigmentary abnormalities.

Contact allergy patch testing

Contact allergy patch testing is an important and valuable tool for the diagnosis of suspected allergic

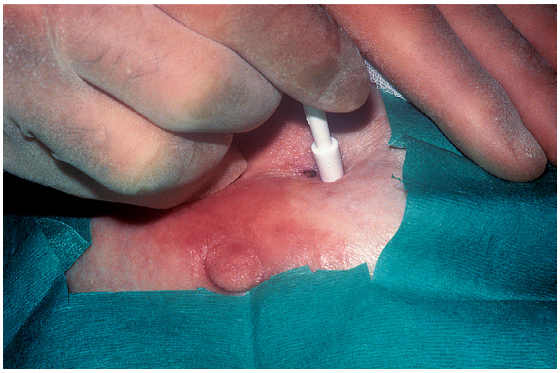


Figure 20.31 Disposable punch skin biopsy, especially useful where minimal scarring is desirable.

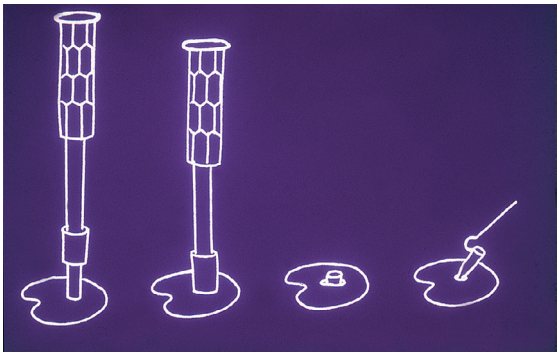


Figure 20.32 A 4-mm punch diagnostic biopsy.

dermatitis owing to contact (contact dermatitis). The formulation of the allergens is critical, and various standard contact allergen test batteries have been developed in different countries and clinics to include the most common culprits. Patch testing is simple, but the results are not always easy to interpret. Allergens are placed in shallow aluminium 1-cm^2 wells and applied in strips to the patient's back for 48 hours for initial reading and a second reading at 96 hours. This ensures that any delayed-type hypersensitivity (e.g. Coomb's type IV reaction to an allergen) can be identified.

Skin biopsy

Biopsy of the skin is used not only to excise benign and malignant tumours but also to identify the nature of reactive and/or inflammatory lesions. Punch biopsy (Figs 20.31 and 20.32) is popular because of its convenience and its minimal scarring. The biopsy can be studied by conventional histology, often supplemented by immunofluorescence and sometimes immunohistochemistry or molecular biology, to identify specific proteins or genetic abnormalities. Skin biopsy of fresh tissue can be sent for the microbiology of unusual infections and is increasingly used in diagnosis and in assessing the progress of skin diseases.